

UYAR, Russian Federation

WMO: 295760

Lat: 55.8192N

Lon: 94.3143E

Elev: 1232

StdP: 14.05

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.6	-29.4	-40.7	0.6	-36.0	-34.3	0.8	-28.9	25.7	19.8	23.0	17.9	3.8	40	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	83.0	64.9	80.0	63.6	76.9	62.4	67.2	78.3	65.7	76.0	64.2	73.6	5.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	92.1	71.2	62.0	86.9	69.4	60.3	81.9	68.7	32.5	78.4	31.3	76.2	30.1	73.5	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	14.3	12.7	-37.4	89.3	7.6	1.9	-42.9	90.7	-47.3	91.8	-51.6	92.8	-57.1	94.2
			-36.6	69.6	7.2	1.5	-41.8	70.6	-46.0	71.5	-50.0	72.3	-55.2	73.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-0.3	8.3	21.5	37.1	48.6	61.1	64.4	60.5	46.3	32.6	13.4	7.5
	DBStd	24.87	17.49	13.81	12.30	8.51	9.71	7.66	5.01	6.56	7.97	9.59	15.55	15.41
	HDD50	7293	1558	1168	885	396	150	9	0	4	165	542	1098	1318
	HDD65	11617	2023	1588	1350	838	516	160	72	171	562	1006	1548	1783
	CDD50	1283	0	0	0	8	106	341	445	329	53	1	0	0
	CDD65	132	0	0	0	0	6	42	53	31	0	0	0	0
	CDH74	1325	0	0	0	3	91	503	429	289	10	0	0	0
	CDH80	248	0	0	0	0	15	102	72	59	0	0	0	0
Wind	WSAvg	5.3	5.2	5.1	5.7	6.5	6.2	4.6	4.2	4.1	4.6	5.5	5.8	5.8
Precipitation	PrecAvg	18.9	0.7	0.6	0.7	1.2	1.7	2.0	2.5	3.5	1.9	1.5	1.4	1.2
	PrecMax	22.8	1.2	1.0	2.2	2.5	3.5	3.2	6.3	7.6	2.8	2.3	2.2	2.3
	PrecMin	13.5	0.2	0.3	0.2	0.3	0.7	0.5	0.3	1.8	0.3	0.3	0.7	0.4
	PrecStd	2.7	0.3	0.2	0.5	0.6	0.8	0.6	1.5	1.5	0.8	0.6	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.3	49.7	67.8	81.9	87.1	86.9	85.3	74.9	62.6	47.3	36.7
		MCWB	29.8	30.9	38.8	48.5	59.7	65.0	67.4	67.2	58.7	48.6	40.3	31.7
	2%	DB	31.3	33.2	44.8	61.4	75.9	82.9	82.1	80.7	68.1	54.6	41.3	32.1
		MCWB	27.0	27.8	36.7	46.2	56.6	64.0	65.8	65.3	56.5	45.2	35.6	28.2
	5%	DB	27.3	30.4	41.0	55.9	70.4	79.8	79.1	76.7	63.2	49.9	37.6	29.2
		MCWB	23.8	26.3	34.0	43.9	53.8	62.8	64.5	63.1	54.2	43.0	32.8	26.6
	10%	DB	23.1	27.5	37.4	51.8	66.2	76.4	75.8	72.9	58.5	46.0	33.5	26.5
		MCWB	20.4	24.2	31.6	41.7	52.3	61.3	63.3	61.9	51.4	40.6	30.3	24.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	31.3	40.2	50.6	61.1	67.9	69.3	69.3	59.3	49.5	41.0	32.5
		MCDB	34.2	36.2	47.3	63.7	78.2	81.1	81.0	82.0	71.4	58.1	46.4	36.4
	2%	WB	26.8	28.8	37.1	47.6	58.2	66.0	67.5	66.6	57.2	46.4	36.2	28.7
		MCDB	30.5	32.3	44.1	58.8	72.9	79.1	77.9	77.2	66.9	53.0	41.0	31.6
	5%	WB	24.1	26.6	34.6	44.8	55.7	64.4	66.1	64.7	54.9	43.8	33.2	26.6
		MCDB	27.7	30.1	40.4	54.6	67.7	76.2	75.5	73.7	61.7	49.1	37.2	29.4
	10%	WB	20.5	24.0	32.1	42.0	53.1	62.6	64.6	62.8	52.6	40.9	30.5	23.9
		MCDB	23.3	27.7	36.7	51.1	64.0	73.7	73.2	71.2	57.7	45.6	33.3	26.2
Mean Daily Temperature Range	5% DB	MDBR	13.1	15.1	16.3	16.5	18.8	19.6	17.0	17.5	15.5	12.9	12.6	12.3
		MCDBR	15.9	15.2	18.2	23.8	26.4	26.5	22.9	23.8	24.6	21.5	13.8	14.0
		MCWBR	12.8	11.9	12.1	13.5	12.9	10.8	8.9	10.2	12.9	13.2	9.9	11.9
	5% WB	MCDBR	14.7	14.4	16.7	22.0	22.6	23.4	18.6	21.0	22.2	18.7	13.7	13.4
		MCWBR	11.8	11.6	11.5	13.5	11.6	10.0	8.1	9.5	12.1	12.3	10.5	11.8
Clear-Sky Solar Irradiance	taub		0.275	0.308	0.351	0.387	0.406	0.401	0.420	0.402	0.360	0.340	0.308	0.276
	taud		2.034	2.016	1.985	2.144	2.201	2.307	2.271	2.288	2.360	2.262	2.082	1.977
	Ebn at Noon		191	231	250	262	265	267	259	255	251	221	174	152
	Edh at Noon		21	31	42	42	42	38	39	36	29	25	20	17
All-Sky Solar Radiation	RadAvg		258	577	1027	1378	1576	1883	1698	1374	886	484	245	159
	RadStd		26	34	71	80	157	159	125	117	105	37	30	16

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page